

2024

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

Compulsory

1. Choose the correct option from the following multiple choice questions :

$$2 \times 10 = 20$$

(i) What is the primary goal of Black Box Testing?

- a) To validate the internal code structure
- b) To find security vulnerabilities
- c) To verify the functionality of the software
- d) To review the source code

(ii) Which of the following refers to a different set of tasks that ensure that the software that has been built is traceable to customer requirements?

- a) Verification
- b) Validation
- c) Array
- d) None of these

(iii) In which of the following categories can White Box testing be classified?

- a) Error guessing technique
- b) Structural testing
- c) Design based testing
- d) None of these

(iv) Alpha testing is performed by :

- a) End Users
- b) Developers
- c) Testers within the organizations
- d) Automated tools

(v) Regression testing is done to:

- a) Check performance
- b) Verify old features work after changes
- c) Improve design
- d) Ensure code is readable

(vi) The process of finding and fixing defects is called:

- a) Debugging
- b) Reviewing
- c) Re-testing
- d) Validation

(vii) Which testing is done without executing the code?

- a) Dynamic Testing
- b) Static Testing
- c) Integration Testing
- d) Unit Testing

(viii) Boundary value analysis is related to:

- a) Black-box testing
- b) White-box testing
- c) Unit testing
- d) Performance testing

(ix) What is a test case?

- a) A document describing coding steps
- b) A document containing test steps and expected output
- c) A backup file of the software
- d) A report of errors

(x) A higher cyclomatic complexity implies :

- a) Simpler code
- b) Easier testing
- c) More complex and harder to maintain code
- d) Fewer bugs

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

2. Explain Verification and Validation with examples.
3. What is Cyclomatic Complexity? How is it calculated?
4. What is black box testing?
5. What is boundary value analysis? Give an example.
6. Why we use decision tables?
7. What are test metrics? Describe different types of software test metrics and their significance.

Section-C

Long Answer type Questions

Answer any **two** questions of the following :

$$15 \times 2 = 30$$

8. Explain different levels of software testing with suitable examples.

9. What is V-Shaped software life cycle model? Explain its advantages and disadvantages.
10. Explain control flow testing and key components of control flow testing, and how is it used to design test cases? Illustrate with an example.
11. What is coverage and what are the different types of coverage technique?

